SSPI RTL testing report:

10 different variations of the SSPI RTL were designed for testing the SSPI's behaviour with existing testcases in trisoc and gc2025 and a custom sspi testing setup.This setup was based on an existing I2S testbench setup adapted to work with SSPI, made for quick compilation and testing with waveform debugging support.

All ten RTL test variants differ from default SSPI only inside rl_receive_idle_to_data_receive and/or by the presence of a new rule rl_set_up_time_simplex_rx dedicated for simplex_rx mode of operation.

The variants differ in this rule's guard:

```
rule rl_receive_idle_to_data_receive(
     wr_write_en == 1                                    // (A) write-edge clause
  && rx_fifo.enqReadyN(1) && rg_receive_state == IDLE
  && rg_total_bit_rx != 0
  && (rg_comm_mode != 2'b10 || rg_transmit_state == TRANSMIT_DONE)
  && rg_ncs == 0 && rg_spi_en == 1
  && (rg_comm_mode != 2'b01 || rg_set_up_count == rg_cs_t_delay));     // (B) setup-count
clause
```

And some variants have this rule immediately after rl_set_up_time:

```
rule rl_set_up_time_simplex_rx(rg_receive_state == IDLE && rg_total_bit_rx != 0
     && rg_comm_mode == 2'b01 && rg_ncs == 0 && rg_spi_en == 1
     && rg_set_up_count < rg_cs_t_delay);
   rg_set_up_count <= rg_set_up_count + 1;
   rg_active <= SETUP_PHASE;
endrule
```

RTL variants:
1. default - 26/32
Default SSPI RTL used in devices/spi_v2 by Shakti. No rl_set_up_time_simplex_rx rule.

2. first_edge - 16/32
In rl_receive_idle_to_data_receive, clause (A) becomes CPHA-selected and clause (B) is deleted:
((wr_read_en == 1 && rg_clk_phase == 0) || (wr_write_en == 1 && rg_clk_phase == 1))
No rl_set_up_time_simplex_rx rule.

3. no_edge - 20/32
In rl_receive_idle_to_data_receive, clause (A) is deleted outright and clause (B) is deleted, so the rule fires on any cycle the FIFO has room and rg_ncs == 0. No rl_set_up_time_simplex_rx rule.

4. no_edge_rx - 20/32
In rl_receive_idle_to_data_receive, clause (A) gains a receive-only exemption and clause (B) is deleted:
(wr_write_en == 1 || rg_comm_mode == 2'b01)
No rl_set_up_time_simplex_rx rule.

5. setup_rx - 28/32
rl_receive_idle_to_data_receive is unchanged from default. Adds rl_set_up_time_simplex_rx with the guard shown above, ungated by any strobe.

6. setup_rx_no_edge - 20/32
Adds the ungated rl_set_up_time_simplex_rx. In rl_receive_idle_to_data_receive, clause (A) is deleted outright; clause (B) is retained.

7. setup_rx_no_edge_rx - 32/32 (MOST SUCCESSFUL)
Adds the ungated rl_set_up_time_simplex_rx. In rl_receive_idle_to_data_receive, clause (A) becomes (wr_write_en == 1 || rg_comm_mode == 2'b01); clause (B) is retained. This is the only variant passing all 32 cases.

8. setup_rx_rd_en - 12/32
Adds rl_set_up_time_simplex_rx with an extra wr_read_en == 1 term in its guard. In rl_receive_idle_to_data_receive, clause (A) changes strobe from wr_write_en == 1 to wr_read_en == 1; clause (B) is retained.

9. setup_rx_wr_en - 26/32
Adds rl_set_up_time_simplex_rx with an extra wr_write_en == 1 term in its guard. rl_receive_idle_to_data_receive is unchanged from default.

10. setup_rx_wr_en_no_edge_rx - 30/32
Adds rl_set_up_time_simplex_rx with an extra wr_write_en == 1 term in its guard. In rl_receive_idle_to_data_receive, clause (A) becomes (wr_write_en == 1 || rg_comm_mode == 2'b01); clause (B) is retained.

Table of results for each RTL variant covering all 32 testcases:

```
============================== SETUP TIME = 0 ==============================
RTL VARIANT               |   simplex   | full_duplex |  duplex_msb |  duplex_lsb | score
                          | ll lh hl hh| ll lh hl hh| ll lh hl hh| ll lh hl hh|
--------------------------+------------+------------+------------+------------+------
default                   | F  P  F  P | P  P  P  P | P  P  P  P | P  P  P  P | 14/16
first_edge                | F  P  F  P | W  P  W  P | W  P  W  P | W  P  W  P |  8/16
no_edge                   | P  P  P  P | P  P  P  P | P  P  P  P | P  P  P  P | 16/16
no_edge_rx                | P  P  P  P | P  P  P  P | P  P  P  P | P  P  P  P | 16/16
setup_rx                  | F  P  F  P | P  P  P  P | P  P  P  P | P  P  P  P | 14/16
setup_rx_no_edge          | P  P  P  P | P  P  P  P | P  P  P  P | P  P  P  P | 16/16
setup_rx_no_edge_rx       | P  P  P  P | P  P  P  P | P  P  P  P | P  P  P  P | 16/16
setup_rx_wr_en            | F  P  F  P | P  P  P  P | P  P  P  P | P  P  P  P | 14/16
setup_rx_wr_en_no_edge_rx | P  P  P  P | P  P  P  P | P  P  P  P | P  P  P  P | 16/16
setup_rx_rd_en            | F  F  F  F | W  W  W  W | W  W  W  W | W  W  W  W |  0/16

============================== SETUP TIME = 1 ==============================
RTL VARIANT               |   simplex   | full_duplex |  duplex_msb |  duplex_lsb | score
                          | ll lh hl hh| ll lh hl hh| ll lh hl hh| ll lh hl hh|
--------------------------+------------+------------+------------+------------+------
default                   | W  W  W  W | P  P  P  P | P  P  P  P | P  P  P  P | 12/16
first_edge                | F  P  F  P | P  W  P  W | P  W  P  W | P  W  P  W |  8/16
no_edge                   | P  P  P  P | W  W  W  W | W  W  W  W | W  W  W  W |  4/16
no_edge_rx                | P  P  P  P | W  W  W  W | W  W  W  W | W  W  W  W |  4/16
setup_rx                  | F  P  F  P | P  P  P  P | P  P  P  P | P  P  P  P | 14/16
setup_rx_no_edge          | P  P  P  P | W  W  W  W | W  W  W  W | W  W  W  W |  4/16
setup_rx_no_edge_rx       | P  P  P  P | P  P  P  P | P  P  P  P | P  P  P  P | 16/16
setup_rx_wr_en            | F  F  F  F | P  P  P  P | P  P  P  P | P  P  P  P | 12/16
setup_rx_wr_en_no_edge_rx | F  P  F  P | P  P  P  P | P  P  P  P | P  P  P  P | 14/16
setup_rx_rd_en            | F  F  F  F | P  P  P  P | P  P  P  P | P  P  P  P | 12/16

Key:  P = pass    W = ran to completion, wrong data    F = hang / no data
Cols: ll/lh/hl/hh = CPOL/CPHA  (l=0, h=1)
```